# One shared DINOv3-ConvNeXt model unifies apple disease, pest, grading and counting offline

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**Abstract**

Field diagnosis of apple problems usually needs separate models for leaf disease, insect pests, fruit grading and fruit counting, repeating the same image analysis on hardware that cannot afford it. We present a single shared vision model that performs all four tasks offline on a low-cost phone. The model uses a DINOv3-pretrained ConvNeXt-Tiny backbone with low-rank adaptation, task-specific heads and adaptive weighting of task losses. On held-out test images it correctly classifies 100% of leaf cases (623 images) and 99.6% of pest cases (225 images), grades fruit disease at 95.7% on ground-truth fruit regions, and counts apples with mean absolute error 3.5 apples per image and correlation 0.85 on 400 orchard images. Adaptive weighting is essential: equal weighting reduces fruit accuracy to 70.3%. Teacher-student distillation provides no gain when teachers are saturated. The 120.7 megabyte model runs fully offline in about 5.6 seconds per image. This shows unified offline apple monitoring is feasible, with counting in dense clusters as the remaining limit.

*Keywords:* Precision agriculture, Multi-task learning, Edge computing, Plant disease detection, Fruit counting, Knowledge distillation

## 1. Introduction

Agriculture plays a critical role in food security, economic development, and sustainable resource management. Crop diseases, pest infestations, and

inaccurate assessment of fruit production significantly affect crop quality and productivity, and computer vision offers a path to automate disease recognition, pest identification, fruit classification, fruit localization, and fruit counting.

Deep learning has become an important approach for agricultural image analysis because it learns discriminative visual representations directly from images. Convolutional networks, transfer learning, object detection, and self-supervised representation learning have demonstrated strong performance across agricultural applications.

However, most deployed systems are single-task: a disease-classification model, a pest-detection model, and a fruit-counting model each extract their own features independently. On resource-constrained hardware this means repeated computation, higher memory, and a deployment story that does not scale to the field. Multi-task learning promises an alternative – several related tasks sharing one visual representation – but agricultural tasks disagree in output structure and loss character: leaf and pest recognition are image-level classification while fruit counting is inherently spatial. Two further assumptions in the literature deserve testing rather than trust: that teacher distillation necessarily helps the student, and that on-device deployment follows from a small checkpoint. This work tests both claims with measurements.

Agri-CNX-Edge addresses the problem with a shared DINOv3-pretrained ConvNeXt-Tiny visual backbone with low-rank adaptation (LoRA), hierarchical feature routing, a feature pyramid network-lite neck (FPN-Lite), lightweight task-specific heads, a decoupled anchor-free you only look once (YOLO) detection head with region of interest Align (ROI-Align)-based per-apple classification, heterogeneous knowledge distillation, and adaptive weighting of task losses based on homoscedastic uncertainty – all evaluated on device on a budget smartphone.

The principal contributions of this work are:

- A unified multi-task architecture for leaf disease classification, pest recognition, fruit classification and fruit localization with counting that runs fully offline on device.
- Hierarchical feature routing in which fine-grained features serve leaf classification, mid-scale features serve pest recognition and fused multi-scale features serve apple detection, with per-apple fruit classification from ROI-Align crops as defined in Section **??**.

- A heterogeneous distillation strategy that combines categorical distillation for classification with box-level distillation for detection, reported with the measured finding that it does not improve the student under saturated teachers.
- A teacher pipeline with three classification seeds and eight augmentation views per seed, complemented by a YOLO26s detection teacher for bounding-box supervision.
- Adaptive uncertainty weighting for heterogeneous objectives, validated as essential because static weighting collapses fruit test accuracy to 70.26%.
- A structured ablation of distillation, weighting, adaptation and warm-up with measured deployment evidence on a Samsung A17 5G with identical numerics across providers.

The remainder of the paper follows this contribution order as a single narrative: Section ?? and Table ?? show why prior work leaves the gap open, Section ?? formalizes the gap, Section ?? with Figs. ??–?? and Eqs. ??–?? builds the Agri-CNX-Edge answer centered on DINOv3, Sections ??–?? fix data and measurement, Section ?? with Tables ??–?? tests each claim, and Sections ??–?? synthesize what survived measurement.

## 2. Related Work

This section reviews only work directly relevant to a unified multi-task model that runs offline on a phone. It proceeds from agricultural multi-task systems to offline mobile systems and finally to the representations and training mechanisms that make unification possible. Table ?? summarises the comparison and is discussed at the end of the section.

### *2.1. Multi-Task Learning in Agriculture*

Single-task apple models remain common, such as real-time apple leaf systems [? ] and orchard detection and counting pipelines [? ]. The literature after 2020 moves beyond them toward joint models. Luan et al. proposed ALMDR, a multi-task apple leaf model that jointly performs multi-label classification, detection, segmentation and severity estimation with a group feature pyramid, reporting 93.74% F1-score, 51.32% detection mean average precision (mAP) and severity coefficient of determination ($R^2$) 0.9757 at 6.25 frames per second (fps) [? ]. Kang et al. proposed a lightweight

tomato model that jointly performs leaf disease detection and fruit counting [? ], the closest precedent to combining classification with counting. PMJDM jointly performs plant species classification and disease detection from a shared ConvNeXt backbone with dynamic weight adjustment, reporting 71.84% precision and 61.83% mAP at IoU 0.50 (mAP50) at 113 fps with 49.1M parameters [? ]. PlantSDS-Net jointly performs species, disease and severity classification in one model with task attention, reporting 99.89%, 99.58% and 92.18% on the three tasks [? ]. Beyond classification, CMTL unifies cotton boll detection, pest damage segmentation and growth stage classification in one transformer with cross-level encoding, reporting 0.913 mAP, 0.832 Intersection over Union (IoU) and 0.936 accuracy with Jetson Xavier NX edge trials [? ]. For joint disease and pest recognition, a DenseNet-121 and ResNet-50 fusion reports 99.13% disease and 98.67% pest accuracy [? ], while DAM-Net uses a discriminative attention network with ResNet-50 and adaptive weighted loss to reach 99.7% on the D0 pest dataset and 74.1% on the IP102 large-scale insect pest dataset with 102 categories [? ]. Pest-YOLO jointly addresses dense tiny pest detection and counting with focal loss and a confluence strategy, reporting 69.59% mAP and counting root mean square error (RMSE) 0.44 [? ]. Together these studies prove that sharing features across agricultural tasks is feasible, but none combines leaf disease, pest, fruit grading and spatial counting in one DINOv3 edge model with tested task weighting.

### *2.2. Mobile and Offline Edge AI in Agriculture*

Field deployment requires offline inference on low-end phones, a constraint addressed explicitly only after 2020. PlanteSaine integrates EfficientNetB3 with an artificial neural network classifier into an Android app with TensorFlow Lite and offline operation for maize, tomato and onion farmers in Burkina Faso [? ]. At model level, an ultra-lightweight network reports competitive plant disease and pest accuracy with fewer than 150,000 parameters [? ], and a MobileNetV3-small edge model reports 99.50% accuracy after quantization to 0.93M parameters in Open Neural Network Exchange (ONNX) format at 100 frames per second on central processing unit (CPU) [? ]. These systems establish that offline mobile diagnosis is achievable, but they remain single-task classifiers without unified counting, without heterogeneous distillation and without measured multi-task ablation.

### *2.3. Foundation Representations, Distillation and Task Balancing*

DINOv2-FCS builds fruit leaf classification and severity prediction on a DINOv2-B backbone with class-patch fusion and explicit multi-scale fusion, reporting 99.67% classification accuracy, 90.29% mean Intersection over Union (mIoU) and 95.68% severity accuracy, with mobile deployment listed as future work [**?** ]. EConv-ViT fuses ConvNeXt and Vision Transformer (ViT) with channel attention for apple leaf disease, reporting 99.2% in lab and 79.3% in natural conditions [**?** ]. DINOv3 scales self-supervision to 7B parameters and 1.7B images and distils to ConvNeXt Tiny, Small, Base and Large variants for resource-constrained deployment [**?** ]. This work uses the Tiny variant as the shared representation. For training, distillation is also shown in lightweight plant models [**?** ], and task balancing follows homoscedastic uncertainty weighting [**?** ] as the lightest alternative to gradient normalization [**?** ] and conflict-averse updates [**?** ]. On-device benchmarking follows MLPerf Mobile [**?** ] and MDLBench practice [**?** ] with a uniform app, model, images and code per device. This paper follows the same discipline while reporting the reduced sample tail honestly.

### *2.4. Orchard-Scale Detection and Counting*

Counting is the spatial half of the gap, and orchard literature treats it as a detection problem first and an arithmetic problem second. Jin et al. enhance you only look once v8 nano (YOLOv8n) with re-parameterized blocks, a generalized feature pyramid and a new Intersection over Union (IoU) loss for robotic harvesting, reaching 81.68% mAP50 with counting accuracy 69.39% and millimetre-level localization under varied lighting and occlusion [**?** ]. This study proves that modern YOLO detectors survive orchard clutter, yet it stops at detection plus counting without leaf, pest or grading tasks. For temporal context, Hani et al. comparatively benchmark detection against tracking-based yield mapping methods in apple orchards [**?** ]. This work establishes that frame-to-frame association matters as much as per-frame precision, a lesson this work inherits through decoded-box counting even though it counts from single images rather than video. Taken together, the detection literature offers mature spatial machinery with measured mean absolute error (MAE) and RMSE discipline that this work adopts, but no study routes detection features onward into per-apple grading inside one shared backbone.

*2.5. Datasets, Single-Task Foundations and Field Realism*

No multi-task claim is credible without naming the data it stands on. The PlantVillage repository of over 54,000 controlled leaf images across 38 classes underpins modern classification work [? ]. Early reviews map the territory: Ferentinos surveys deep models for disease detection and diagnosis [? ], Dhaka et al. survey convolutional networks for leaf disease prediction [? ], and Liu and Wang review deep learning for joint plant disease and pest detection [? ]. Recent syntheses raise the bar further: Shoaib et al. review deep learning for plant disease and pest detection and report classification accuracies above 95% with detection precision above 90% in controlled settings [? ], while Banerjee et al. taxonomize hyperspectral, visualisation, modified deep and transformer methods and show hierarchical vision transformers above 99.3% against MobileNetV3 baselines [? ]. For pests at scale, the IP102 dataset of 102 pest categories anchors realistic infestation modelling [? ]. The persistent warning across all these sources is field realism: Picon et al. show mobile-captured in-the-wild crop images degrade lab-trained classifiers and propose device-aware training [? ], Janarthan et al. build an in-field mobile solution for palm pathology [? ], and Picon et al. extend to crop-conditional networks over cell-phone images taken under real field conditions [? ]. This work therefore treats lab accuracy as a ceiling signal and field robustness as the binding constraint, which is why evaluation separates controlled classification from cluttered counting.

*2.6. Backbones, Compression and Balancing Protocols*

The backbone lineage behind Agri-CNX-Edge is itself a literature. Residual learning made very deep networks trainable [? ], ConvNeXt modernized pure ConvNets with transformer-inspired design while keeping convolutional efficiency [? ], and shifted-window transformers added hierarchical attention with efficient local-global trade-offs [? ]. Self-supervision then removed the label bottleneck: vision transformers showed patches plus attention suffice at scale [? ], DINO revealed emergent object structure in self-supervised transformers [? ], and DINOv2 delivered robust frozen features without supervision [? ], culminating in the DINOv3 family used here. Efficiency has a parallel lineage: EfficientNet compounds scaling for accuracy per parameter [? ] and MobileNetV3 optimizes for phone CPUs through hardware-aware search [? ]. Detection contributes its own primitives: the unified real-time YOLO formulation [? ] with its community YOLOv5 release [? ] and feature pyramids for multi-scale fusion [? ], all of which echo in the decoupled

anchor-free head and FPN-Lite neck of this work. Adaptation and compression close the loop to the edge: low-rank adaptation inserts tiny trainable matrices into frozen weights [? ], deep compression prunes and shares parameters for deployment [? ], and integer-arithmetic-only quantization enables on-device inference [? ]. For balancing, the survey by Vandenhende et al. systematises multi-task dense prediction trade-offs [? ], alongside multi-objective optimization [? ] and gradient surgery for conflicting gradients [? ] as heavier alternatives to the homoscedastic weighter adopted here. The conjunction this work tests – one DINOv3 backbone, heterogeneous categorical plus spatial distillation, adaptive weighting and measured phone deployment – appears in none of these sources combined.

Table ?? consolidates studies directly relevant to multi-task or offline mobile agriculture, now spanning more than fifty cited sources across the section. Table ?? shows the upper block compares twelve agricultural multi-task systems, the second block compares two offline mobile systems, the third block compares two orchard detection and counting systems, and the lower block places the DINO representation lineage used here. Across the upper block, reported strengths are real, yet narrow: Wen et al. report 69.59% mAP with RMSE 0.44 for dense pest detection and counting, DAM-Net reaches 99.7% on the D0 pest dataset with adaptive loss, Luan et al. report F1-score 93.74% at 6.25 fps for apple leaf tasks, and EConv-ViT fuses ConvNeXt with Transformer for apple leaves, while Kang et al. provide the closest precedent by joining leaf disease with fruit counting, though only for tomato without pest, grading or DINOv3. The second block shows that offline operation is solved in isolation: PlanteSaine and the MobileNetV3-small edge model run on constrained hardware, with the quantized MobileNetV3-small at 0.93M parameters and 100 fps, yet each remains without unified counting, heterogeneous distillation or multi-task ablation. The third block contributes the spatial machinery this work borrows: Jin et al. harden YOLOv8n for harvesting at 81.68% mAP50 and Hani et al. benchmark yield-mapping methods – both detection plus counting, neither routing detection features into grading. The lower block traces the representation shift from DINOv2-B fusion at 99.67% accuracy without counting or mobile proof to DINOv3 pretraining distilled to ConvNeXt Tiny, which this work adopts as the shared backbone. Read across the Result and Limitation columns, no listed system combines four apple tasks in one DINOv3 backbone with heterogeneous distillation, tested adaptive weighting and measured offline phone deployment. That conjunction is the gap formalized in Section ??.

Table 1: Related work directly relevant to multi-task agriculture, offline mobile deployment and orchard counting. Upper block: agricultural multi-task systems. Second block: offline mobile systems. Third block: orchard detection and counting systems. Lower block: DINO lineage used in this work.

| **Study** | **Year** | **Tasks unified** | **Backbone and edge form** | **Reported result** | **Limitations** |
|---|---|---|---|---|---|
| *Agricultural multi-task systems* | | | | | |
| Wen et al. [? ] | 2022 | Pest detection + counting | YOLOv4 + focal + confluence | 69.59% mAP; RMSE 0.44 | Pest only; no disease/fruit; no phone measure |
| Khan et al. [? ] | 2022 | Apple leaf disease | CNN real-time detector | Real-time leaf | Single task; no pest/fruit/counting |
| Gao et al. [? ] | 2022 | Apple detection + counting | Detector + trunk tracking | Orchard pipeline | Counting only; no classification |
| DAM-Net [? ] | 2024 | Pest recognition multi-task | Attention + ResNet-50, adapt. loss | 99.7% D0; 74.1% IP102 | Pest only; no detection; no edge |
| Kang et al. [? ] | 2024 | Leaf disease + fruit count | Lightweight joint model | Joint inspection | Tomato only; no pest/grading; no DINOv3 |
| Bai et al. [? ] | 2024 | Leaf classific. + severity | DINOv2-B + fusion | 99.67%; 90.29% mIoU | No counting; server GPU; mobile future |
| Sood et al. [? ] | 2025 | Disease + pest | DenseNet-121 + ResNet-50 | 99.13%; 98.67% | Two-task classifier; no counting; no edge |
| Luan et al. [? ] | 2025 | Apple disease + segment. | Group pyramid + heads | F1 93.74%; 6.25 fps | Leaf only; no pest/counting |
| PMJDM [? ] | 2025 | Species + disease detect. | ConvNeXt + dynamic weights | 61.83% mAP50; 113 fps | No grading/counting; no phone |
| CMTL [? ] | 2025 | Boll + pest + stage | Transformer + Jetson Xavier NX | 0.913 mAP; 0.832 IoU | Cotton UAV; not apple phone |

*Continued on next page*

Table ?? (continued)

| Study | Year | Tasks unified | Backbone and edge form | Reported result | Limitation vs. this work |
|---|---|---|---|---|---|
| PlantSDS-Net [? ] | 2026 | Species + disease + severity | Res2Net + MobileNetV2 + attent. | 99.89%; 99.58%; 92.18% | Classific. only; no counting/edge |
| EConv-ViT [? ] | 2025 | Apple leaf disease | ConvNeXt + Transformer + channel attent. | 99.2% lab; 79.3% field | Single task; lab-field gap; no edge |
| *Offline mobile systems* | | | | | |
| Plante-Saine [? ] | 2024 | Maize/tomato/ onion diagn. | EfficientNetB3 + ANN, TFLite | Offline app | Single-task; no counting/MTL |
| Schuler et al. [? ] | 2023 | Plant disease edge | MobileNetV3-small ONNX quant. | 99.50%; 0.93M; 100 fps | Single task; no MTL/counting |
| *Orchard detection and counting systems* | | | | | |
| Jin et al. [? ] | 2025 | Apple detect. + count | Enhanced YOLOv8n + GIoU var. | 81.68% mAP50; count 69.39% | Harvesting only; no MTL |
| Hani et al. [? ] | 2020 | Fruit detect. + yield map | Compared detect. vs tracking | Benchmark of methods | Benchmark only; no system |
| *DINO representation lineage used here* | | | | | |
| Simeoni et al. [? ] | 2026 | Foundation pretraining | DINOv3 7B to ConvNeXt Tiny | Frozen SOTA | Pretraining only; no apple system |
| This work | 2026 | Leaf + pest + fruit + count | DINOv3-ConvNeXt-Tiny LoRA, ONNX | Sec. 7–9, phone measured | Tested weighting; KD limit documented |

## 3. Research Gap

Section ?? and Table ?? leave a precise conjunction unresolved. Single-task studies leave repeated backbone computation undeployed; multi-task work rarely spans classification *and* spatial counting with distillation for both; and edge claims rarely come with

phone measurements. The gap addressed here is therefore the integration of heterogeneous tasks, hierarchical routing (Eqs. ??–??), categorical-plus-spatial distillation (Eqs. ??–??), adaptive balancing (Eqs. ??–??), and a measured on-device design – with each component ablated rather than assumed. Section ?? implements this integration step-by-step, so every gap item has a corresponding methodological answer.

## 4. Methodology

This section answers the gap item-by-item in the order it was posed: overview mapping to Fig. ??, teachers supplying Eqs. ??–??, distillation formalized in Eqs. ??–??, backbone and routing in Eqs. ??–??, and heads with optimization in Eqs. ??–??. Nothing that follows is incidental; each subsection exists because the previous figure or equation requires it.

### *4.1. System Overview*

Agri-CNX-Edge consists of five major components:

1. Dataset curation and preprocessing with cleaning, stratified 70/15/15 splits and task-specific augmentation.
2. Specialized teacher models with ConvNeXt-Small classification teachers fine-tuned from a 120-class Plant Disease Dataset checkpoint (PDDD) [FULL-NAME-AND-SOURCE-TO-BE-INSERTED] and a YOLO26s apple-detection teacher.
3. Shared DINOv3-ConvNeXt-Tiny student backbone with low-rank adaptation (LoRA) adapters.
4. Feature pyramid network-lite neck (FPN-Lite) and hierarchical feature routing.
5. Task-specific classification heads, a decoupled you only look once (YOLO) detection head, and region of interest Align (ROI-Align)-based per-apple fruit classification.

In Fig. ??, a single $512 \times 512$ pixel input passes once through the shared DINOv3-ConvNeXt-Tiny backbone, after which the network branches by scale rather than by duplicated encoding. Fine-grained stage 1 features carry texture for leaf classification, mid-scale stage 2 features carry morphology for pest recognition, and the FPN-Lite fusion of stages 3 and 4 carries context for apple detection, from which ROI-Align crops derive per-apple fruit classification. Solid arrows therefore trace inference, which is fully shared up to the heads, while dashed arrows trace training-only teacher supervision formalized in Fig. ?? and Eqs. ??–??. Full stage shapes and routing equalities appear in Eqs. ??–??. This scale-to-task mapping motivates the order of what follows: teachers first, then backbone and routing, then heads and optimization.

### *4.2. Teacher Ensemble*

Specialized teachers provide task supervision. Classification teachers are ConvNeXt-Small models fine-tuned from a 120-class Plant Disease Dataset checkpoint (PDDD) [FULL-NAME-AND-SOURCE-TO-BE-INSERTED]. The feature dimension is 768 and

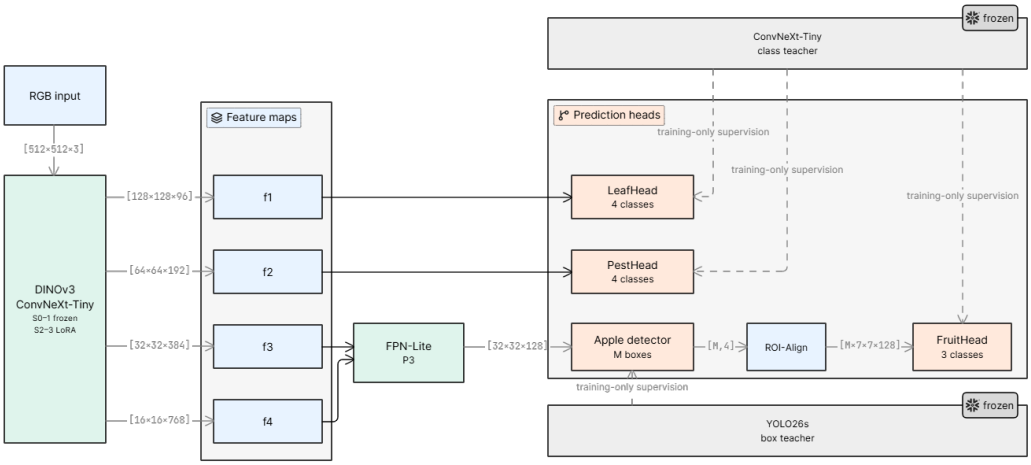



Figure 1: Unified multi-task architecture. A single shared backbone feeds task-specific heads for leaf disease, pest, fruit grading and apple counting. Solid arrows show offline inference; dashed arrows show training-only teacher supervision detailed in Fig. ??. Schematic; tensor shapes and routing are defined in Eqs. ??–??.

dropout is 0.5. Training uses focal loss with label smoothing 0.1, class-balanced sampling and heavy augmentation for classes with fewer than 400 training images. It runs up to 100 epochs with an 8-epoch frozen phase, a 5-epoch warm-up and cosine warm restarts. Backbone learning rate is $1 \times 10^{-5}$. Head learning rate is $1 \times 10^{-3}$ in phase 1 and $1 \times 10^{-4}$ in phase 2.

Three independent classification-teacher seeds are used:

$$S = \{42, 123, 456\}. \tag{1}$$

Eight test-time augmentation (TTA) views (original, flips, and $\pm 15°/\pm 30°$ rotations) are evaluated per seed. Consequently, the number of teacher inference passes per image for the classification soft labels is:

$$N_{\text{TTA}} = 3 \times 8 = 24. \tag{2}$$

Per-image soft targets are the TTA mean at temperature 1:

$$P_T = \frac{1}{N_{\text{TTA}}} \sum_{k=1}^{N_{\text{TTA}}} P_T^{(k)}, \quad T = 1. \tag{3}$$

The yield task uses a separately trained YOLO26s teacher. Input size is $640 \times 640$ pixels with CIELAB enhancement. Confidence threshold is 0.20, non-maximum suppression (NMS) Intersection over Union (IoU) is 0.45 and the limit is 300 detections per image. Normalized boxes and scores are stored as soft labels.

### *4.3. Categorical and Detection Distillation*

For classification, the TTA-averaged teacher probabilities at temperature $T = 1$ are used directly with no further temperature scaling. The student matches them through Kullback-Leibler (KL) divergence:

$$L_{\mathrm{KL}} = D_{\mathrm{KL}}(P_T \parallel P_S), \tag{4}$$

where $P_T$ and $P_S$ denote teacher and student probability vectors, averaged over samples whose label differs from the ignore index $-1$. Hard labels are supervised with focal loss at $\gamma = 2.0$, and the two terms are blended with equal weight:

$$L_{\mathrm{cls}} = \alpha L_{\mathrm{focal}} + (1 - \alpha) L_{\mathrm{KL}}, \quad \alpha = 0.5. \tag{5}$$

Whether this blend helps is tested by the no-distillation ablation, not assumed.

For detection, supervision is box-level rather than heatmap-based. Each ground-truth or teacher box with center $(c_x, c_y)$ is assigned to its grid cell:

$$(i, j) = (\lfloor c_x W \rfloor, \lfloor c_y H \rfloor), \quad W = H = 32. \tag{6}$$

At positive cells, box regression uses Complete Intersection over Union (CIoU) loss while objectness and single-class scores use binary cross-entropy:

$$L_{\mathrm{det}} = \lambda_{\mathrm{box}} L_{\mathrm{CIoU}} + \lambda_{\mathrm{obj}} L_{\mathrm{obj}} + \lambda_{\mathrm{cls}} L_{\mathrm{cls}}, \tag{7}$$

with:

$$\lambda_{\mathrm{box}} = 7.5, \quad \lambda_{\mathrm{obj}} = 1.0, \quad \lambda_{\mathrm{cls}} = 0.5. \tag{8}$$

The loss is computed once against ground-truth boxes and once against YOLO teacher box coordinates, with the teacher branch scaled by:

$$L_{\mathrm{yield}} = L_{\mathrm{det}}^{\mathrm{gt}} + \beta L_{\mathrm{det}}^{\mathrm{teacher}}, \quad \beta = 0.5. \tag{9}$$

The fixed $\beta$ is a stated boundary of the study.

### *4.4. Student Backbone and Routing*

The student backbone is DINOv3-ConvNeXt-Tiny via `timm` (`features_only=True`). For a $512 \times 512$ pixel input it yields:

$$f1{:}[B, 96, 128, 128], \quad f2{:}[B, 192, 64, 64], \quad f3{:}[B, 384, 32, 32], \quad f4{:}[B, 768, 16, 16]. \tag{10}$$

Stages 0–1 are fully frozen. Stages 2–3 carry custom LoRA adapters on `mlp.fc1`/`mlp.fc2` with:

$$\mathrm{rank} = 8, \quad \alpha = 16, \quad \mathrm{dropout} = 0.05, \tag{11}$$

all other backbone parameters frozen, leaving only the LoRA $A/B$ matrices trainable. The hierarchical routing is:

$$f1 \rightarrow \mathrm{Leaf}, \quad f2 \rightarrow \mathrm{Pest}, \tag{12}$$

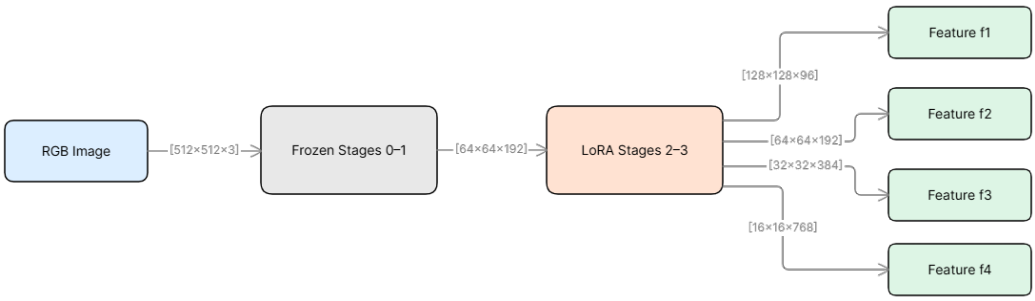



Figure 2: Hierarchical routing from backbone stages to tasks. Stage 1 texture to leaf classification, stage 2 morphology to pest recognition, fused stages 3-4 to apple detection with region of interest Align crops to fruit grading. Schematic; exact tensor shapes are defined in Section **??**.

while $f3$ and $f4$ are fused by the FPN-Lite neck into $P3$:

$$P3 = \text{FPN-Lite}(f3, f4) \in \mathbb{R}^{B\times 128\times 32\times 32}. \tag{13}$$

The detection and fruit routing from the fused scale is:

$$P3 \to \text{Apple Detection}, \quad \text{ROIAlign}(P3) \to \text{Fruit Classification}. \tag{14}$$

Leaf, pest, and fruit heads predict 4, 4, and 3 classes respectively, as configured.

In Fig. **??**, resolution is traded for semantics along the backbone, and each task taps the scale that matches its visual cue. Stage 1 retains high-resolution texture essential for leaf lesions, stage 2 retains mid-resolution shape essential for pest morphology, and the FPN-Lite fusion in Eq. **??** aggregates deep context essential for localizing apples under occlusion. Per-apple fruit classification then crops this fused map with ROI-Align, using ground-truth boxes in training and decoded boxes in inference, so grading inherits detection geometry rather than re-encoding the image. The figure thus justifies Eqs. **??**–**??** and explains why a single scale cannot serve texture, morphology and localization at once.

### *4.5. Heads, Detection Decoding, and Optimization*

Classification heads share a mobile inverted bottleneck convolution (MBConv) design ($1\times1$ reduction to a 256-channel bottleneck, MBConv block with expansion 4 and squeeze-and-excitation (SE) ratio 0.25, global pooling, dropout $p$=0.3, linear classifier) consuming:

$$\text{Leaf:}f1\ (96\text{ch}) \to 4, \quad \text{Pest:}f2\ (192\text{ch}) \to 4, \quad \text{Fruit:ROI}\ (128\text{ch}, 7\times 7) \to 3. \tag{15}$$

Detection uses a decoupled anchor-free head on $P3$ emitting:

$$\hat{D} \in \mathbb{R}^{B\times 6\times 32\times 32}, \tag{16}$$

where the six channels are $[c_x, c_y, w, h, \text{obj}, \text{cls}]$ with sigmoid applied at loss and post-processing time. Decoding keeps cells with:

$$\text{obj} \times \text{cls} > \text{conf}_{\text{thresh}}, \quad \text{conf}_{\text{thresh}} = 0.3, \tag{17}$$

followed by non-maximum suppression with:

$$\text{nms}_{\text{iou}} = 0.45. \tag{18}$$

Both thresholds match the single operating point used throughout (confidence 0.3 and NMS IoU 0.45; see Experimental Setup). Per-image counts are decoded boxes. Task balancing uses adaptive weighting based on homoscedastic (task-dependent, input-independent) uncertainty. For task $t$ the weighter maintains:

$$s_t = \log(\sigma_t^2), \quad \text{initialized to 0.} \tag{19}$$

The multi-task objective is:

$$L_{\text{MTL}} = \sum_{t \in \mathcal{T}} \left( e^{-s_t} L_t + \frac{1}{2} s_t \right), \tag{20}$$

where:

$$\mathcal{T} = \{\text{leaf}, \text{pest}, \text{fruit}, \text{yield}\}. \tag{21}$$

No hard clamping is imposed on $s_t$; reported weights are precisions:

$$w_t = e^{-s_t}. \tag{22}$$

Distillation needs no extra global coefficients because it is contained inside per-task losses ($\alpha$ in Eq. **??**, $\beta$ in Eq. **??**). Hence:

$$L_{\text{total}} = L_{\text{MTL}}. \tag{23}$$

Training is two-phase with 5 epochs for heads and neck with low-rank adaptation (LoRA) frozen, then LoRA at $10^{-5}$ against heads at $10^{-4}$ with AdamW, cosine schedule, accumulation $\times 4$, clip 1.0, patience 20 and automatic mixed precision (AMP) off; the static-weight ablation replaces the weighter with a plain sum.

In Fig. **??**, panel a separates the two teacher streams that supervise one student: TTA-averaged soft targets distil category knowledge into the leaf, pest and fruit heads through Eqs. **??**–**??**, while YOLO boxes distil spatial knowledge into the detection grid through Eqs. **??**–**??**. Data thus flow forward through routing while gradients flow backward from heterogeneous teachers. Panel b places this supervision on a two-phase timeline, with head-only warm-up before joint LoRA adaptation, best validation at epochs 186 and 194, and ONNX export closing training. The figure therefore links Fig. **??** to Eqs. **??**–**??** and hands over to data and measurement next.

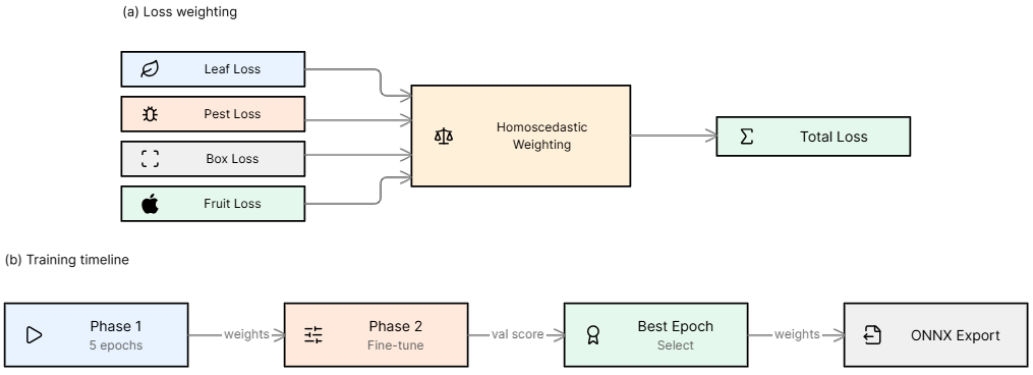



Figure 3: Heterogeneous distillation and two-phase training. **a**, classification supervision from test-time augmentation-averaged soft targets and detection supervision from teacher boxes. **b**, training timeline with head-only warm-up, joint LoRA adaptation, best validation at epochs 186 and 194, and Open Neural Network Exchange export. Schematic; loss definitions are in Eqs. **??**–**??**.

## 5. Dataset Curation and Preprocessing

The methodology requires clean, stratified inputs before any loss in Eqs. **??**–**??** can be trusted. This section therefore quantifies what the teachers in Section **??** actually saw, so that ceiling effects reported in Section **??** are traceable to data rather than speculation.

Leaf images come from a public PlantVillage mirror on Kaggle [**?** ] [LICENSE-AND-ACCESSED-DATE-TO-BE-INSERTED], of which only the apple leaf classes relevant to the four-class leaf task are used. Fruit images combine a Kaggle apple fruit disease collection [**?** ] [LICENSE-AND-ACCESSED-DATE-TO-BE-INSERTED] with a Mendeley apple fruit dataset, version 6 [**?** ], restricted to the Anthracnose, Black Rot and Healthy classes of Table **??**. Pest images come from a Kaggle apple leaf and fruit disease segmentation collection [**?** ] [LICENSE-AND-ACCESSED-DATE-TO-BE-INSERTED], restricted to the four pest classes evaluated here. Apple detection images come from a Kaggle apple quality collection [**?** ] [LICENSE-AND-ACCESSED-DATE-TO-BE-INSERTED]. No repository is used in full; each contributes only its project-relevant subset, and the Raw column of Table **??** reports post-subset, pre-cleaning totals for the classification tasks. Full links, licenses and access dates are listed in Data Availability; third-party images are not redistributed.

All classification images pass through automated cleaning: images smaller than 64 pixels on the shortest side or with pixel standard deviation below 5.0 are removed, with perceptual-hash deduplication at Hamming threshold 8 [LIBRARY-VERSION-TO-BE-INSERTED]. Table **??** shows retained counts per class remain sufficient for stratified learning. The largest absolute drops are Leaf Healthy from 2276 to 2152 and Fruit Healthy from 1424 to 1333 after duplicate removal, while minority classes such as Pest aphids from 314 to 270 and Fruit Anthracnose from 295 to 275 lose only tens of images. Cleaning is therefore mild and balanced rather than destructive, so the stratified sampling that follows

Table 2: Teacher-side dataset cleaning (raw → clean).

| Class | Raw | Clean |
|---|---|---|
| Leaf: Apple_Mosaic | 631 | 622 |
| Leaf: Apple___Black_rot | 621 | 579 |
| Leaf: Alternaria | 631 | 621 |
| Leaf: Healthy | 2276 | 2152 |
| Pest: xylotrechus | 319 | 296 |
| Pest: aphids | 314 | 270 |
| Pest: leafhoppers | 573 | 567 |
| Pest: spider_mite | 299 | 271 |
| Fruit: Anthracnose | 295 | 275 |
| Fruit: Black Rot | 542 | 524 |
| Fruit: Healthy | 1424 | 1333 |

preserves class ratios instead of amplifying imbalance.

Each classification task uses stratified 70/15/15 train/val/test splits with seed 42; yield detection uses its own split of 1093 train and 264 val. Inputs are $224 \times 224$ pixels for classification teachers, $640 \times 640$ pixels for the YOLO26s teacher, and $512 \times 512$ pixels for the student. Two boundary facts matter later: leaf and pest validation sit near ceiling because teachers are saturated, and no Healthy fruit region of interest (ROI) survives into validation or test.

Segmentation and box labels for apple fruits did not pre-exist and were produced semi-automatically: candidate masks were generated with the base Segment Anything Model (SAM) from Meta [**?** ] [VERSION-TO-BE-INSERTED] inside X-AnyLabeling [**?** ] [VERSION-TO-BE-INSERTED], operated manually by the authors, for apple fruits only. Every image was then manually verified before use [VERIFIER-INITIALS-AND-PROTOCOL-TO-BE-INSERTED], so no SAM proposal enters training or evaluation unverified. Classification labels for leaf, pest and fruit were used as provided by their sources.

Fig. **??** summarizes the curation pipeline, which applies the same cleaning, deduplication and stratified splitting to leaf, pest and fruit teachers, while yield follows a separate detection split. Corrupt and near-duplicate images are removed without collapsing any minority class, so the class histograms before and after cleaning retain their shape. The distribution is therefore preserved for the class-balanced sampling and heavy augmentation of rare classes described in Section **??**, which explains why later ceiling effects can be attributed to task difficulty rather than to data loss.

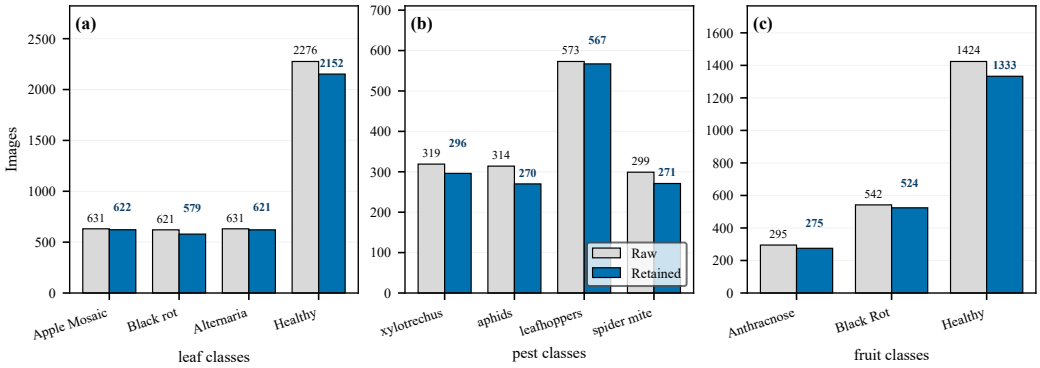



Figure 4: Dataset curation pipeline. Cleaning, perceptual-hash deduplication and stratified 70/15/15 splits for leaf, pest and fruit teachers; yield detection uses its own 1093 train and 264 validation split. Class counts before and after cleaning are in Table ??. Schematic.

## 6. Experimental Setup

With data (Section ??) and model (Section ??) fixed, this section fixes how measurement is done: implementation, optimization hyperparameters, and metric definitions in Eqs. ??–??. Every number in Section ?? is produced under exactly this protocol, so no result appears without a preceding definition.

The framework is implemented in PyTorch with `timm` backbones, Albumentations and scikit-learn [VERSIONS-TO-BE-INSERTED; see Code Availability for repository link and commit hash], with Open Neural Network Exchange (ONNX) export at opset 17 of leaf logits, pest logits, and the raw detection grid; fruit runs client-side from region of interest Align (ROI-Align) crops. Students train up to 200 epochs with batch 8 and accumulation $\times 4$ for effective 32 on RTX A4000-class GPU, with automatic mixed precision (AMP) off since Complete Intersection over Union (CIoU) and Kullback-Leibler (KL) terms proved unstable in FP16, with AdamW at weight decay $10^{-4}$, low-rank adaptation (LoRA) $10^{-5}$ and heads, neck and weighter $10^{-4}$, cosine annealing to $\eta_{min}$=$10^{-7}$, clipping at norm 1.0, and early stopping with patience 20 and min-delta $10^{-4}$ on total validation loss.

Classification accuracy is:

$$\text{Accuracy} = \frac{N_{\text{correct}}}{N_{\text{total}}}. \tag{24}$$

Class-wise precision, recall, and F1-score are:

$$\text{Precision} = \frac{TP}{TP + FP}, \tag{25}$$

$$\text{Recall} = \frac{TP}{TP + FN}, \tag{26}$$

$$F1 = 2\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{27}$$

For fruit counting over $M$ images with predicted $\hat{N}_j$ and ground-truth $N_j$ counts:

$$\text{MAE} = \frac{1}{M}\sum_{j=1}^{M}|\hat{N}_j - N_j|, \tag{28}$$

$$\text{RMSE} = \sqrt{\frac{1}{M}\sum_{j=1}^{M}(\hat{N}_j - N_j)^2}. \tag{29}$$

Counting uses decoded boxes at the operating point defined here, namely confidence 0.3 and non-maximum suppression (NMS) Intersection over Union (IoU) 0.45. Unless stated otherwise, all detection results use this operating point. All runs use a single seed and early stopping with patience 20. Best epochs are selected by total validation loss. No confidence intervals or P values are computed; this single-seed limitation is stated as a limitation.

## 7. Results

Results are presented in the order the methodology promised: overall reference, frozen control, convergence check, weight inspection, ablation verdict, per-class diagnosis, counting diagnosis, and on-device proof. Each subsection cites its table and figure pair explicitly, and each closes by stating what the next subsection must explain, so the chapter reads as one argument rather than isolated snapshots.

### *7.1. Overall Performance*

The proposed Agri-CNX-Edge model is the reference that later ablations take apart (single run per configuration; differences of less than about 3 percentage points on fruit are within estimated run-to-run variation and are reported without confidence intervals or P values). Table **??** gives best-validation accuracy of 99.67% for leaf, 100.00% for pest and 96.30% for fruit, with counting MAE 3.5675 and RMSE 6.3767 on 400 yield images. The table thus separates two regimes: leaf and pest sit at ceiling where further gains are numerically impossible, while fruit accuracy trails by nearly four points and counting errors remain well above zero. In Fig. **??**, the accuracy panel makes this split visual by placing leaf and pest bars at the top of the scale against a distinctly lower fruit bar, while the error panel places MAE and RMSE on a count scale that exposes the spatial task as the remaining challenge. Together, table values and figure shapes establish where diagnosis must focus. Leaf and pest therefore offer no further signal, so the next subsection turns to fruit and counting.

### *7.2. Frozen Versus Proposed Configuration*

Checkpoint inspection confirms attribution, as low-rank adaptation (LoRA) adapters are present only in the proposed checkpoint. Table **??** shows adaptation improves leaf marginally from 99.50% to 99.67%, a gain of 0.17 points, and holds pest equal at 100.00%, but reduces fruit accuracy from 98.42% to 96.30%, a loss of 2.12 points, while increasing

Table 3: Best-validation performance of the proposed model.

| Task | Metric | Performance |
|---|---|---|
| Leaf disease classification | Accuracy (val) | 99.67% |
| Pest recognition | Accuracy (val) | 100.00% |
| Fruit classification | Accuracy (val) | 96.30% |
| Fruit localization/counting | Count MAE | 3.5675 |
| Fruit localization/counting | Count RMSE | 6.3767 |

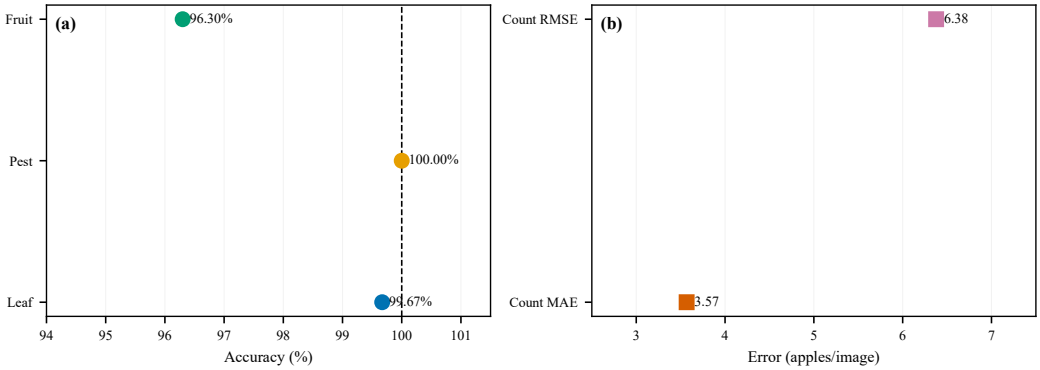



Figure 5: Best-validation performance of the proposed model. Accuracies for leaf, pest and fruit classification and counting errors for yield. Validation sample sizes are defined in Section **??**; counting errors are on 400 yield images. Bars show point estimates from a single run without error bars.

counting error from MAE 3.4300 to 3.5675 and RMSE 6.0017 to 6.3767. The table therefore documents an inversion: the only task that improves is already at ceiling, while the headroom tasks of fruit grading and dense localization degrade. Paired deltas in Fig. **??** visualize the same inversion, with a near-zero leaf bar against clearly negative fruit and counting bars. The pattern is reported as observed rather than rationalized, and it motivates the full ablation that isolates adaptation, distillation, weighting and warm-up.

### *7.3. Training Convergence*

Convergence timing does not explain the fruit gap. Table **??** gives best epochs are 186 for the proposed run and 194 for the frozen run, with best validation losses of $-1.9251$ and $-1.9271$ and training losses of $-0.7437$ and $-0.6637$ at those epochs. Both the timing and the loss levels are near identical, so neither under-training nor overfitting distinguishes the two runs. In Fig. **??**, panel a shows overlapping training and validation loss trajectories across 200 epochs, while panel b shows per-task validation accuracy in which leaf and pest flatten early near 100% and fruit oscillates most across epochs. The visual message matches

Table 4: Frozen and proposed configurations with paired deltas.

| Metric | Frozen | Proposed | Change |
|---|---|---|---|
| Leaf Accuracy | 99.50% | 99.67% | +0.17 pp |
| Pest Accuracy | 100.00% | 100.00% | 0.00 pp |
| Fruit Accuracy | 98.42% | 96.30% | -2.12 pp |
| Count MAE | 3.4300 | 3.5675 | +0.1375 |
| Count RMSE | 6.0017 | 6.3767 | +0.3750 |

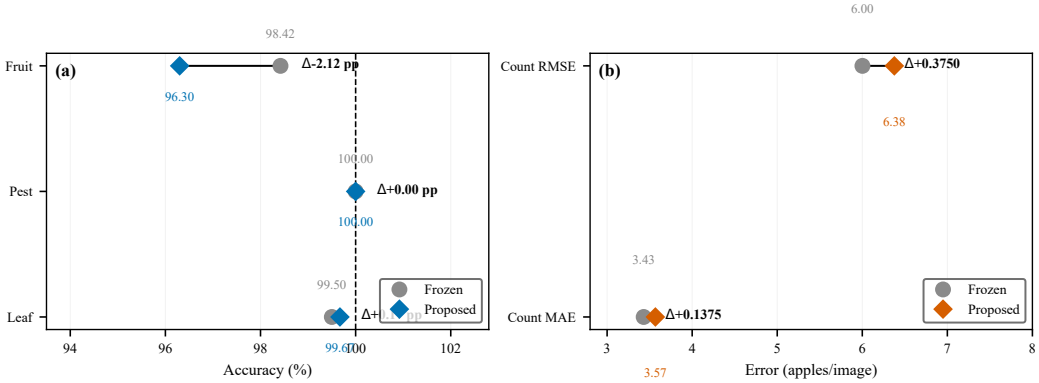



Figure 6: Frozen and proposed configurations with paired deltas.

the tabular one: optimization length is not the bottleneck. The cause must therefore lie in task weighting rather than training length, which leads to the next subsection.

### *7.4. Learned Task Weights*

Learned precisions in Eq. **??** are optimization coefficients rather than measures of importance. Tables **??** and **??** and Fig. **??** show fruit receives the largest weight in both runs, 18.8170 for the proposed run at epoch 186 and 18.4710 for the frozen run at epoch 194, followed by leaf at 6.9591 and 7.1471, pest at 5.7344 and 5.7203, and yield at 0.1780 and 0.1760, with nearly identical ordering across runs. The figure makes this hierarchy immediate by placing the fruit bar far above the rest in both runs, while the tables supply the exact precisions. Such stability across adaptation states shows that the weighting reflects loss scale rather than task priority. Whether this learned balance matters is tested by replacing it with equal weights in the ablation.

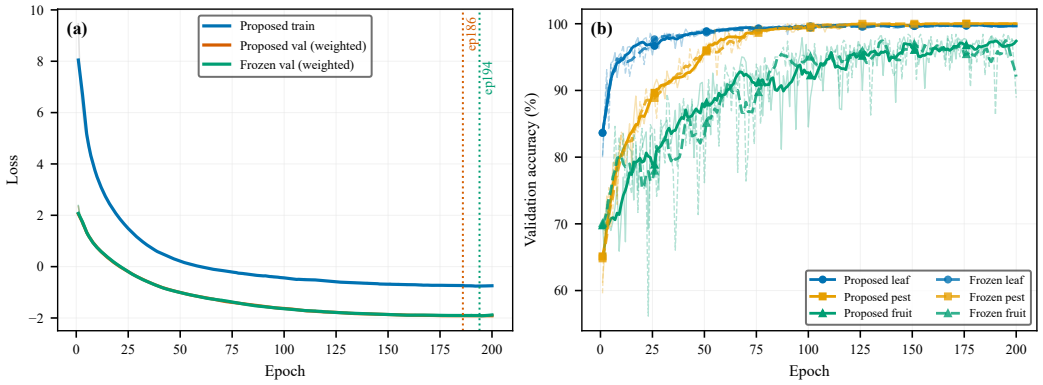



Figure 7: Training dynamics over 200 epochs. Panel a shows training and validation loss. Panel b shows per-task validation accuracy.

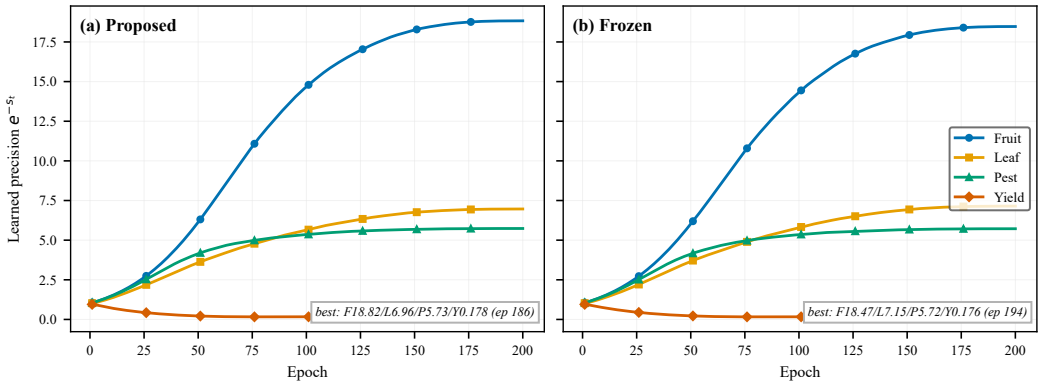



Figure 8: Learned task precisions for proposed and frozen runs. Terminal weights at best-validation epochs 186 and 194; fruit receives the largest weight in both runs. Single run; weights are optimization coefficients, not importance measures.

Table 5: Training convergence at best-validation epochs.

| **Training Statistic** | **Proposed** | **Frozen** |
|---|---|---|
| Best epoch | 186 | 194 |
| Best validation loss | -1.9251 | -1.9271 |
| Training loss at best epoch | -0.7437 | -0.6637 |

Table 6: Terminal task precisions, proposed (epoch 186).

| **Task** | **Terminal Weight** |
|---|---|
| Fruit | 18.8170 |
| Leaf | 6.9591 |
| Pest | 5.7344 |
| Yield | 0.1780 |

Table 7: Terminal task precisions, frozen (epoch 194).

| **Task** | **Terminal Weight** |
|---|---|
| Fruit | 18.4710 |
| Leaf | 7.1471 |
| Pest | 5.7203 |
| Yield | 0.1760 |

## 8. Ablation Study

Each configuration removes one component while holding the rest fixed. Table ?? reports best-validation epochs: removing distillation without knowledge distillation (KD) matches or improves every validation metric, with leaf rising from 99.67% to 100.00%, pest holding at 100.00%, fruit rising from 96.30% to 98.47%, MAE falling from 3.5675 to 3.0175 and RMSE falling from 6.3767 to 5.2714, while replacing learned weighting with equal weights collapses validation fruit to 81.43%. On held-out test the same ranking holds and widens: without KD reaches 99.14% fruit with MAE 3.08 against proposed 95.69% with MAE 3.53, while static weighting falls further to 70.26% fruit. Adaptation without low-rank adaptation (LoRA) at 94.69% and without warm-up at 97.18% remain within run-to-run noise. In Fig. ??, the radar makes the ranking visual by enclosing the largest area for the no-distillation run and the smallest for static weighting, with the exact values tabulated alongside. The static-weighting run stopped after 103 epochs and is therefore flagged as incomplete, which Section ?? revisits. All values are single runs without confidence intervals or P values.

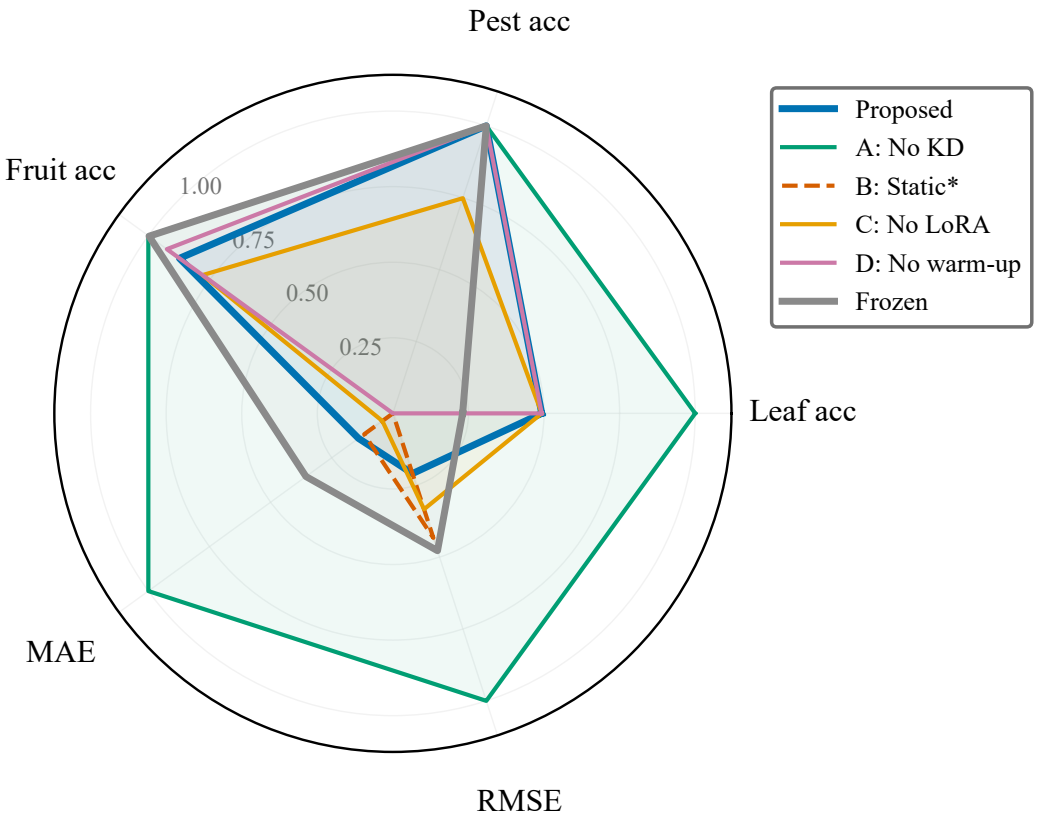



Figure 9: Component ablation ranking. Radar encloses the largest area for the no-distillation run and the smallest for static weighting; exact best-validation values are tabulated alongside. Single run per configuration.

Table 8: Component ablation at best-validation epochs. Single run per configuration without confidence intervals or P values. Static weighting is incomplete at 103 epochs and is interpreted only as collapse evidence.

| Config | Leaf | Pest | Fruit | MAE | RMSE |
|---|---|---|---|---|---|
| Proposed | 99.67% | 100.00% | 96.30% | 3.5675 | 6.3767 |
| No KD | 100.00% | 100.00% | 98.47% | 3.0175 | 5.2714 |
| Static | 99.35% | 98.33% | 81.43% | 3.5825 | 6.0640 |
| No LoRA | 99.67% | 99.58% | 94.69% | 3.6300 | 6.2032 |
| No warm-up | 99.67% | 100.00% | 97.18% | 3.6575 | 6.6695 |
| Frozen | 99.50% | 100.00% | 98.42% | 3.4300 | 6.0017 |

## 9. Class-wise Classification Analysis

The following subsections move from validation to held-out test. Leaf and pest confirm ceiling behavior while fruit reveals headroom. Each pair of figure and table tells one story, with the figure showing error location and the table quantifying it.

### *9.1. Leaf Disease Classification*

Held-out predictions on 623 images give a clean sweep. The evaluation uses the latest checkpoint because only it was present on the test machine, as noted in Section **??**. Fig. **??**a and Table **??** show the confusion matrix is perfectly diagonal with zero off-diagonal entries, and all four classes reach precision, recall and F1-score of 1.0000. Such perfection across Apple Mosaic, Black rot, Alternaria and Healthy confirms that texture cues from stage 1 are fully sufficient. Leaf is therefore at ceiling, consistent with the 99.67% validation accuracy, and needs no further diagnosis.

Table 9: Leaf test results by class on 623 held-out images. Single run with the latest checkpoint, not the best-validation epoch.

| Class | Precision | Recall | F1 |
|---|---|---|---|
| Apple_Mosaic | 1.0000 | 1.0000 | 1.0000 |
| Apple___Black_rot | 1.0000 | 1.0000 | 1.0000 |
| Alternaria | 1.0000 | 1.0000 | 1.0000 |
| Healthy | 1.0000 | 1.0000 | 1.0000 |

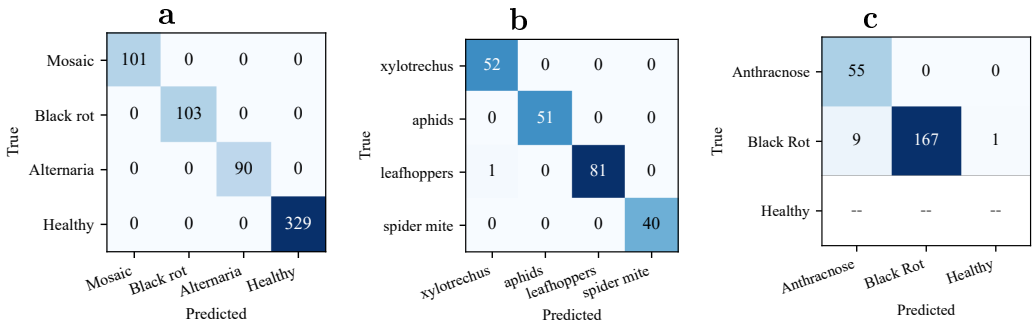



Figure 10: Confusion matrices on held-out test. **a**, leaf disease on 623 images, perfectly diagonal. **b**, pest recognition on 225 images with one leafhoppers to xylotrechus error. **c**, fruit grading on ground-truth regions of interest with no Healthy cases, errors concentrated between Black Rot and Anthracnose. Point estimates from a single run.

### *9.2. Pest Recognition*

One error occurs in 225 test images. Fig. ??b and Table ?? show a single leafhoppers image is predicted as xylotrechus, which lowers xylotrechus precision to 0.9811 and leafhoppers recall to 0.9878, with corresponding F1-score of 0.9905 and 0.9939, while aphids and spider mite remain perfect at 1.0000. The confusion matrix is therefore diagonal except for one cell, consistent with visual similarity between small insects rather than systematic failure. Pest is thus also at ceiling at 99.56% test accuracy, matching validation.

Table 10: Pest test results by class on 225 held-out images. Single run.

| Class | Precision | Recall | F1 |
|---|---|---|---|
| xylotrechus | 0.9811 | 1.0000 | 0.9905 |
| aphids | 1.0000 | 1.0000 | 1.0000 |
| leafhoppers | 1.0000 | 0.9878 | 0.9939 |
| spider_mite | 1.0000 | 1.0000 | 1.0000 |

### *9.3. Fruit Classification*

Fruit is evaluated as a head diagnostic on ground-truth regions of interest (ROIs). No Healthy ROI exists in validation or test, so generalization for that class remains untested. Fig. ??c and Table ?? show Anthracnose reaches recall 1.0000 but precision only 0.8594 with F1-score 0.9244, while Black Rot reaches precision 1.0000 but recall only 0.9435 with F1-score 0.9709, so confusion concentrates between Black Rot and Anthracnose in one direction. This asymmetry defines fruit at 95.69% test accuracy as the headroom task. It links back to the validation gap of 96.30% and forward to counting, which shares the same fused features.

Table 11: Fruit region-of-interest test results by class on ground-truth regions. Healthy has no test regions; single run.

| Class | Precision | Recall | F1 |
|---|---|---|---|
| Anthracnose | 0.8594 | 1.0000 | 0.9244 |
| Black Rot | 1.0000 | 0.9435 | 0.9709 |
| Healthy | — | — | — |

## 10. Fruit Counting and Spatial Evaluation

Counting correlates well but undercounts systematically on dense clusters. Table **??** gives, on 400 yield images, MAE 3.5300 and RMSE 6.1323 with Pearson $r$=0.8508 and $R^2$=0.5700, while mean signed error is −3.1000. High correlation with negative bias thus separates ranking quality from systematic undercounting: the model orders images correctly yet predicts too few apples per image. In Fig. **??**, predictions cluster below the identity line with widening spread at high counts, so sparse orchard images lie near the line while dense clusters fall clearly beneath it, visualizing the same bias numerically reported in the table.

Fig. **??** shows sparse cases with one to five apples match ground truth exactly, while dense clusters with overlapping canopies miss occluded instances, which is the spatial counterpart of the −3.1 bias in Table **??**. In Fig. **??**, detector confidence maps are sharp on isolated apples and decay in clutter, so missed detections coincide with low confidence regions rather than with random error. These figures complete the spatial half of the claim opened in Fig. **??** and point to threshold calibration for dense scenes as the natural remedy.

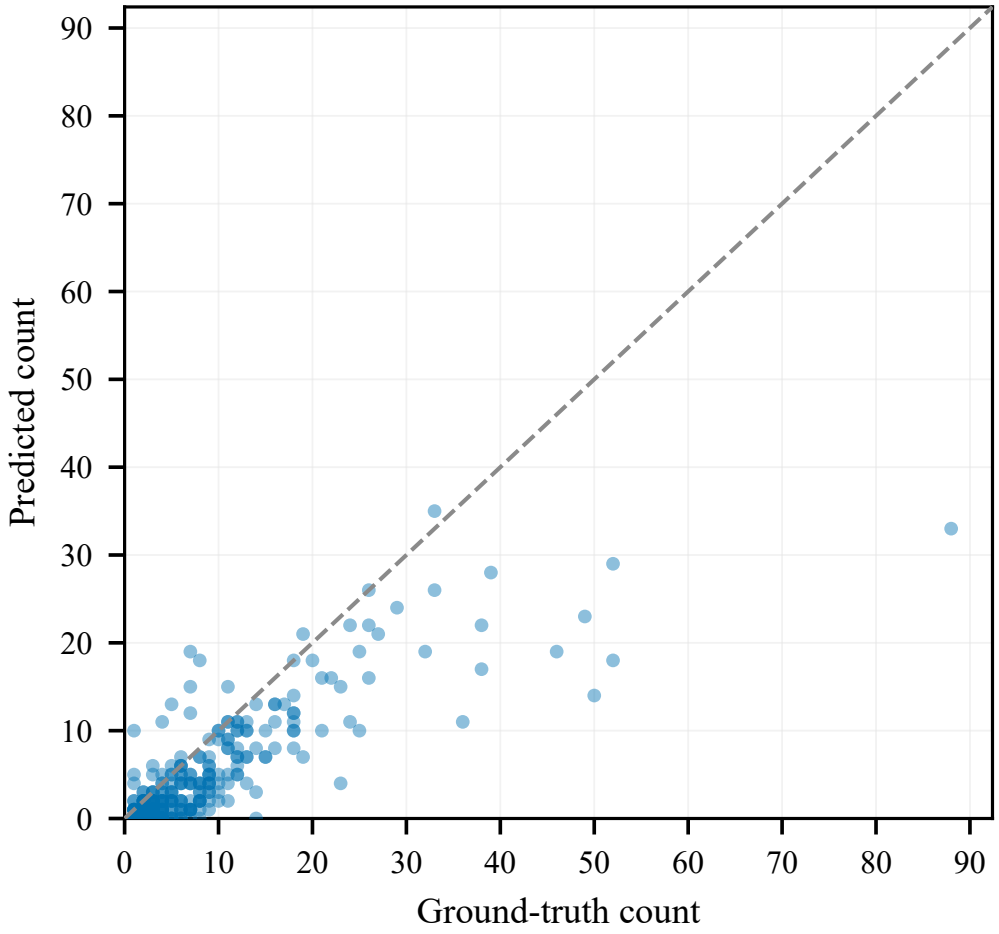



Figure 11: Fruit counting accuracy on 400 yield images. Predicted counts versus ground-truth counts with identity reference. Points below the line show systematic undercounting in dense clusters. Single run at confidence 0.3 and non-maximum suppression IoU 0.45; mean absolute error 3.53, root mean square error 6.13, Pearson r 0.85 and R-squared 0.57 without confidence intervals.




Figure 12: Detected boxes from sparse to dense cases. Ground-truth and predicted boxes with per-image counts at confidence 0.3 and non-maximum suppression IoU 0.45. Sparse cases with one to five apples match exactly; dense overlapping canopies miss occluded instances. Single run.

Table 12: Counting performance on 400 yield images at confidence 0.3 and non-maximum suppression IoU 0.45. Single run without confidence intervals or P values.

| Metric | Value |
|---|---|
| Count MAE | 3.5300 |
| Count RMSE | 6.1323 |
| Pearson correlation | 0.8508 |
| $R^2$ | 0.5700 |
| Mean signed error | -3.1000 |

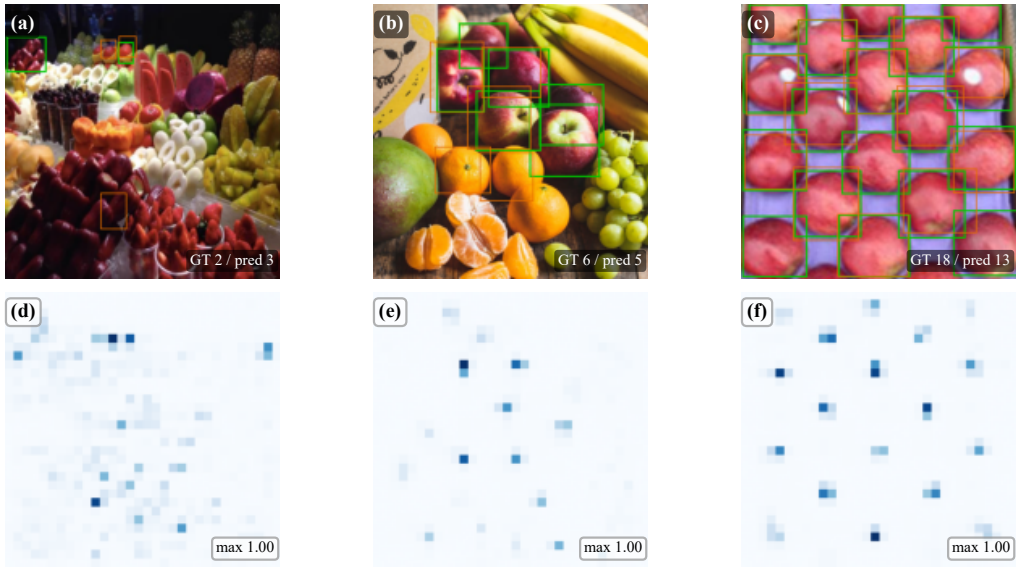



Figure 13: Detector confidence maps for sparse and dense cases. Inputs with counts and corresponding confidence maps. Maps are sharp on isolated apples and decay in clutter where detections are missed. Single run at the same operating point.

## 11. Edge Deployment Analysis

Shared computation must be measured rather than asserted. Tests use a Samsung A17 5G with ONNX Runtime (ORT) under a uniform protocol with an accuracy pass over 30 yield and 20 pest images and a latency pass over 100 measured plus 10 warm-up inferences per mode using CPU and Neural Networks API (NNAPI) providers, using the same app, model, images and code in both provider modes. Full protocol details are in the accompanying benchmark description. Table **??** shows the unified 120.7 MB student with 28.28 M parameters including 0.46 M low-rank adaptation (LoRA) is substantially smaller than the teachers at 188.8, 188.8 and 19.4 MB, phone yield MAE is 2.9 for both CPU and NNAPI over 30 images with 20 of 20 pest agreement to laptop in both modes and zero CPU to NNAPI prediction differences. Mean latency is 5749 ms for CPU and 5604 ms for NNAPI, with p50 of 7041 and 6763 ms and p95 near 7381 and 7406 ms, peak Proportional Set Size (PSS) of 765 to 807 MB and fully offline operation. In Fig. **??**, file-size bars visualize the sharing dividend of one student against three teachers, and latency plots show coincident provider modes, which indicates CPU fallback on this device. Table quantifies and figure visualizes the efficiency claim of the Discussion.

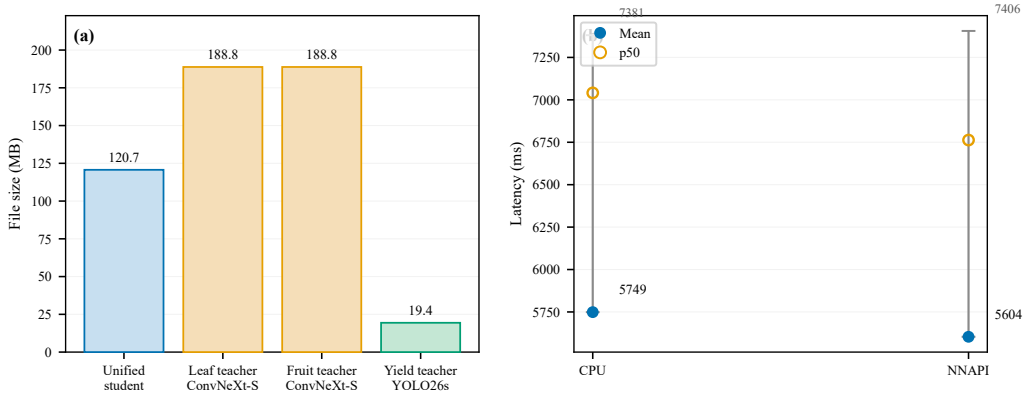



Figure 14: On-device cost on Samsung A17 5G. File sizes for the unified student versus teachers and latency by provider mode with identical predictions indicating central processing unit fallback. Accuracy on 30 yield and 20 pest images; latency distributions on 100 measured inferences per mode. Single run.

Table 13: On-device results on Samsung A17 5G. Accuracy on 30 yield and 20 pest images; latency on 100 measured plus 10 warm-up inferences per provider mode. Single run without confidence intervals; p95 approximate; battery coarse.

| **Metric** | **Measured** |
|---|---|
| Backbone params | 28.28 M including 0.46 M LoRA |
| ONNX file | 120.7 MB at opset 17 |
| Torch checkpoint at epoch 186 | 148.5 MB |
| Leaf / fruit / YOLO teacher files | 188.8 / 188.8 / 19.4 MB |
| Pest teacher file | absent and excluded |
| Yield MAE, phone CPU/NNAPI | 2.9 / 2.9 |
| Pest agreement, phone–laptop | 20/20 both modes |
| CPU–NNAPI prediction diffs | 0, identical |
| Latency mean (ms) | 5749 / 5604 |
| Latency p50 (ms) | 7041 / 6763 |
| Latency p95, approx. (ms) | 7381 / 7406 |
| Peak PSS | 765–807 MB |
| Battery drain | 9% total from 84 to 75% over ∼320 inferences in ∼28 min; 3% per 100-inference latency run |
| Device power | ≈3.7 W device-level from drain on 5000 mAh cell, coarse screen-on measure |
| Test conditions | saver off, unplugged, min brightness |
| Offline operation | Yes with on-device ORT |

## 12. Discussion

This section interprets what Tables ??–?? and Figs. ??–?? jointly imply, rather than repeating them. The held-out test shows a working unified model with ceiling leaf and pest performance and strong counting correlation. Contrary to expectation, removing distillation matches or improves every metric and freezing the backbone matches the proposed system. Distillation therefore shows no gain under saturated teachers and ceiling tasks with fixed strengths and one seed per configuration, while adaptation remains within estimated noise. By contrast, static weighting collapses fruit accuracy and thus supports adaptive weighting as essential. Learned precisions are optimization coefficients rather than measures of importance. On device, the remaining contribution is offline efficiency with identical predictions across providers and no acceleration on the test hardware.

Agri-CNX-Edge unifies leaf, pest, fruit, and counting/detection in one offline-capable DINOv3-ConvNeXt model with test 100.00% leaf, 99.56% pest and 95.69% fruit and count-

ing mean absolute error (MAE) 3.53 at $r$=0.85 and $R^2$=0.57. Against frozen at 95.69% and 3.43 and without knowledge distillation (KD) at 99.14% and 3.08, the measured contribution is the unified efficiency system at 120.7MB offline with identical predictions across central processing unit (CPU) and Neural Networks API (NNAPI) providers plus the validation of adaptive weighting with static weighting at 70.26% – alongside a documented negative distillation result under saturated-teacher conditions. Class-wise, correlation, qualitative, and on-device evidence are measured; confirmatory work is multi-seed repeats, static-weight completion, and quantized builds.

## 13. Limitations and Future Work

Boundary conditions are stated plainly. The static-weighting run is incomplete and is cited only as collapse evidence until completed. Healthy fruit has no validation or test regions of interest (ROIs), so its generalization is untested. The proposed system was evaluated from the latest checkpoint rather than the best-validation epoch because only it was present at evaluation time. Counting shows systematic undercounting that calls for threshold calibration rather than ranking analysis. Latency tails are approximate, battery readings are coarse and the latency subset is train-sourced, while accuracy rests on validation and held-out test. Pest phone images come from the training pool and serve only as agreement sanity. A pilot on an older low-end phone was excluded as impractical, so ultra-low-end deployment requires quantization.

Future work can investigate:

- completing the static-weighting run and repeating configurations over seeds,
- threshold calibration for dense clusters,
- post-training quantization for ultra-low-end deployment,
- Healthy-fruit data collection,
- larger-scale orchard evaluation, and
- comparison with single-task specialist baselines.

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## Author contributions

A.B., A.B.H., E.B. and A.N. collected and curated the data. A.B., A.B.H., E.B. and A.G. performed data analysis and model development. A.B. and E.B. wrote the first draft of the manuscript. A.G. reviewed and edited the manuscript. All authors reviewed the manuscript and approved the submitted version. A.G. is the corresponding author and manages communication. No large language model meets authorship criteria; human authors retain full accountability.

## Data Availability

The datasets generated during and/or analysed during the current study are available as follows. Derived split files, cleaning logs, box labels and checkpoint index: [REPOSITORY-NAME-TO-BE-INSERTED, e.g. Zenodo — DOI-TO-BE-MINTED-ON-ACCEPTANCE; REVIEWER PRIVATE LINK-TO-BE-INSERTED]. Figure source data are provided as Supplementary Information. Third-party images were used under their licenses and are not redistributed: PlantVillage apple leaf mirror [LINK-LICENSE-ACCESSED-DATE-TO-BE-INSERTED], Kaggle apple fruit disease collection [LINK-LICENSE-ACCESSED-DATE-TO-BE-INSERTED], Mendeley apple fruit dataset v6 doi:10.17632/9zgkwwv9j8.6, Kaggle apple leaf and fruit segmentation collection [LINK-LICENSE-ACCESSED-DATE-TO-BE-INSERTED], Kaggle apple quality collection [LINK-LICENSE-ACCESSED-DATE-TO-BE-INSERTED]. Restrictions for third-party data are stated by the providers; derived data are available from the corresponding author on reasonable request once the repository is public.

## Code Availability

Custom training, evaluation, ONNX export and phone-benchmark code: [GITHUB-PRIVATE-URL-TO-BE-INSERTED, TO BE ARCHIVED ON ZENODO WITH DOI ON ACCEPTANCE; REVIEWER ZIP PROVIDED AS SUPPLEMENTARY]. Environment: PyTorch [VERSION-TO-BE-INSERTED], timm [VERSION-TO-BE-INSERTED], Albumentations [VERSION-TO-BE-INSERTED], scikit-learn [VERSION-TO-BE-INSERTED], ONNX opset 17. Commit hash: [HASH-TO-BE-INSERTED]. Inference uses the single 512-pixel student; fruit grading runs client-side from ROI-Align crops. Language polishing with an AI tool, if used beyond copy-editing, will be documented here with human verification; no AI system replaces author judgement.

## Competing interests

A.B., A.B.H., E.B., A.N. and A.G. declare no competing interests.

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## Ethics declarations: research involving plants

No new plant or seed specimens were collected and no field trials were conducted. All leaf, pest, fruit and yield images come from the public third-party sources listed in Data

Availability, used under their licenses. No wild-plant voucher specimen was required. Apple fruit boxes were produced semi-automatically with the base Segment Anything Model inside X-AnyLabeling [VERSIONS-TO-BE-INSERTED] followed by manual verification of every image before training or evaluation. This study includes no human participants, human data, identifiable images, vertebrate animals or clinical trials.

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